

$$16. \text{ Tukey: } W = 5.12 \cdot \sqrt{\frac{41.25}{5}} = 14.91$$

if $|\bar{x}_i - \bar{x}_j| > 14.91$ 則不同

$$① |\bar{x}_1 - \bar{x}_2| = 27.4 > 14.91 \text{ 顯著不同}$$

$$② |\bar{x}_1 - \bar{x}_3| = 10.6 < 14.91 \text{ 不顯著}$$

$$③ |\bar{x}_1 - \bar{x}_4| = 6.4 < 14.91 \text{ 不顯著}$$

$$④ |\bar{x}_2 - \bar{x}_3| = 16.8 > 14.91 \text{ 顯著不同}$$

$$⑤ |\bar{x}_2 - \bar{x}_4| = 33.8 > 14.91 \text{ 顯著不同}$$

$$⑥ |\bar{x}_3 - \bar{x}_4| = 17 > 14.91 \text{ 顯著不同}$$

$$17. (a) H_0: \mu_1 = \mu_2 = \mu_3$$

H_a : 至少一組不同

$$MSTR = \frac{74806}{2} = 37403$$

$$MSE = \frac{225930}{27} = 8367.778$$

$$F = 4.470$$

$$p\text{-value} = 0.02 < 0.05$$

$\therefore$ 拒絕 H_0

$$(b) \bar{x}_1 - \bar{x}_2 = -45$$

$$SE = \sqrt{MSE \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$SE = \sqrt{8367.778 \left(\frac{1}{10} + \frac{1}{10} \right)}$$

$$(c) W = q_{0.05}(3, 27) \sqrt{\frac{8367.778}{10}} \\ = 10.43$$

$\therefore$ Physical Science, Social science
有顯著差異

24. (a) treatments: 4 (stores)

blocks: 8 (items)

$$(b) \text{treatmentss} = 13.1923 \quad df = 3 \quad MS_{\text{Treat}} = \frac{13.1923}{3} = 4.3974$$

$$\text{blockss} = 40.9949 \quad df = 7 \quad MS_{\text{block}} = \frac{40.9949}{7} = 5.8564$$

$$\text{errorss} = 3.4817 \quad df = 21 \quad MSE = \frac{3.4817}{21} = 0.1658$$

$$F = 26.32, p < 0.05$$

$\therefore$ reject H_0

$$(c) F = \frac{5.8564}{0.1658} = 35.32$$

$\rightarrow$ 显著差异

25. (a) $q_{0.05}(k, df)$

$$\Rightarrow q_{0.05}(4, 21) \approx 3.94$$

$$(b) w = 3.94 \cdot \sqrt{\frac{0.1658}{8}} = 0.567$$

$$(c) \textcircled{1} \text{Vons, Ralphs} = 0.205 < 0.567 \rightarrow \text{不显著}$$

$$\textcircled{2} \text{Vons, Stater Bros} = 0.285 < 0.567 \rightarrow \text{不显著}$$

$$\textcircled{3} \text{Vons, Winn-Dixie} = 1.453 > 0.567 \rightarrow \text{显著}$$

$$\textcircled{4} \text{Ralphs vs. Winn-Dixie} = 1.65 > 0.567 \rightarrow \text{显著}$$

$$\textcircled{5} \text{Stater Bros, Winn-Dixie} = 2.171 > 0.567 \rightarrow \text{显著}$$

26. (a) Treatment: 3種 propellant mixtures
blocks: 5位 investigators

$$SS_{\text{Treat}} = 107.2$$

$$SS_{\text{block}} = 176.4$$

$$SS_E = 8.8$$

$$SS_T = 322.4$$

	SS	df	MS	F	p-value
Treatment	107.	2	53.6	11.05	0.00499
Blocks	176.4	4	44.1	9.07	0.00452
Error	38.8	8	4.85		
Total	322.4	14			

(b) $H_0: \mu_1 = \mu_2 = \mu_3$

H_a : 至少1個不同

$$p = 0.0049 < 0.01,$$

$\therefore$ reject H_0

(c) $p = 0.00452 < 0.01$ 差異顯著

$\therefore$ blocking 有效

(d) $w = q_{0.01}(3, 8) \sqrt{\frac{48.5}{5}} = 5.55$

$mix1 \& mix2 < 5.55 \rightarrow$ 不顯著

$mix1 \& mix3 > 5.55 \rightarrow$ 顯著不同

$mix2 \& mix3 < 5.55 \rightarrow$ 不顯著